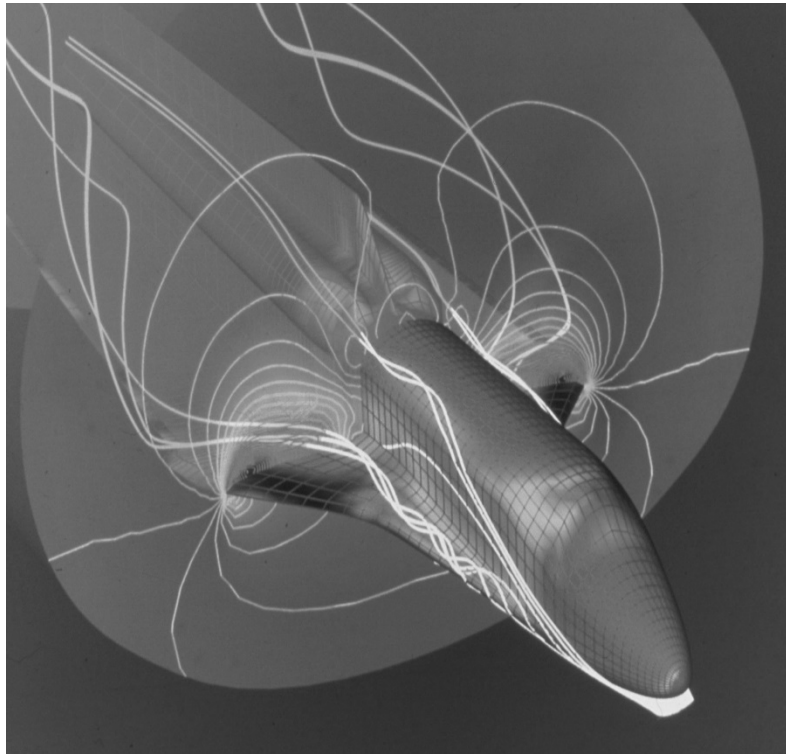


Chapter 4 Differential Relations for Fluid Flow



Learning Objectives

- Recognize the total derivative for fluid in motion.
- Apply the conservation of mass relation for fluid flows.
- Understand Navier-Stokes equations of the linear momentum.
- Apply the conservation of energy equation for fluid flows.
- Recognize boundary conditions for the basic equations.
- Introduce the stream function for the velocity field.
- Understand vorticity and irrotationality in analysis of frictionless irrotational flows.
- Analyze simple incompressible viscous flows including Couette flow and asymmetric flows of cylindrical pipe.

-
- **Total Derivative**
 - **Differential Form of Mass Conservation Equation**
 - **Differential Form of Momentum Conservation Equations**
 - **Differential Form of Energy Conservation Equation**
 - **Boundary Conditions**
 - **Stream Function**
 - **Vorticity, irrotationality**

The Acceleration Field of a Fluid

- The Cartesian vector form of a velocity field is a function of time and space,

$$\mathbf{V}(x, y, z, t) = \mathbf{i}u(x, y, z, t) + \mathbf{j}v(x, y, z, t) + \mathbf{k}w(x, y, z, t)$$

- The **acceleration field** of the flow is defined as the total time derivative of the velocity vector.

$$\mathbf{a}(x, y, z, t) = \frac{d\mathbf{V}}{dt} = \mathbf{i} \frac{du}{dt} + \mathbf{j} \frac{dv}{dt} + \mathbf{k} \frac{dw}{dt}$$

The Acceleration Field of a Fluid

- Applying the chain rule to obtain each scalar time derivative.

$$\frac{du}{dt} = \frac{\partial u}{\partial t} + \frac{\partial u}{\partial x} \frac{dx}{dt} + \frac{\partial u}{\partial y} \frac{dy}{dt} + \frac{\partial u}{\partial z} \frac{dz}{dt} \quad \text{or} \quad a_x = \frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z}$$

- Total derivative of velocity is the acceleration:

$$\mathbf{a}(x, y, z, t) = \frac{d\mathbf{V}}{dt} = \frac{\partial \mathbf{V}}{\partial t} + \left(u \frac{\partial \mathbf{V}}{\partial x} + v \frac{\partial \mathbf{V}}{\partial y} + w \frac{\partial \mathbf{V}}{\partial z} \right) = \frac{\partial \mathbf{V}}{\partial t} + (\mathbf{V} \cdot \nabla) \mathbf{V}$$

Total Acceleration

Local Acceleration

Convective Acceleration

where $\nabla = \mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z}$

- Total derivative operator:

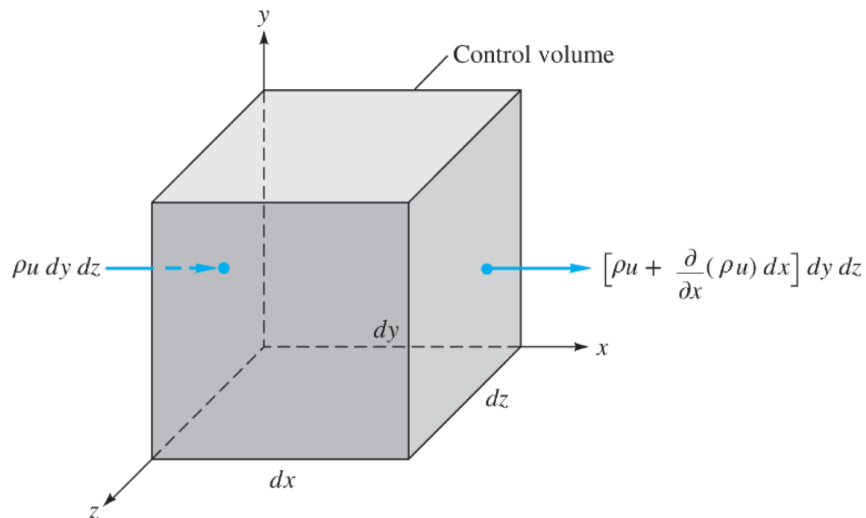
$$\frac{d}{dt} = \frac{\partial}{\partial t} + u \frac{\partial}{\partial x} + v \frac{\partial}{\partial y} + w \frac{\partial}{\partial z} = \frac{\partial}{\partial t} + \mathbf{V} \cdot \nabla$$

The Differential Equation of Mass Conservation

- Conservation of mass, often called the **continuity equation**, states that the fluid mass cannot change.
- The mass changes in the element CV,

$$\int_{CV} \frac{\partial \rho}{\partial t} dV \approx \frac{\partial \rho}{\partial t} dxdydz$$

- Mass flow terms in all six directions,



Face	Inlet mass flow	Outlet mass flow
x	$\rho u dy dz$	$\left[\rho u + \frac{\partial}{\partial x}(\rho u) dx \right] dy dz$
y	$\rho v dx dz$	$\left[\rho v + \frac{\partial}{\partial y}(\rho v) dy \right] dx dz$
z	$\rho w dx dy$	$\left[\rho w + \frac{\partial}{\partial z}(\rho w) dz \right] dx dy$

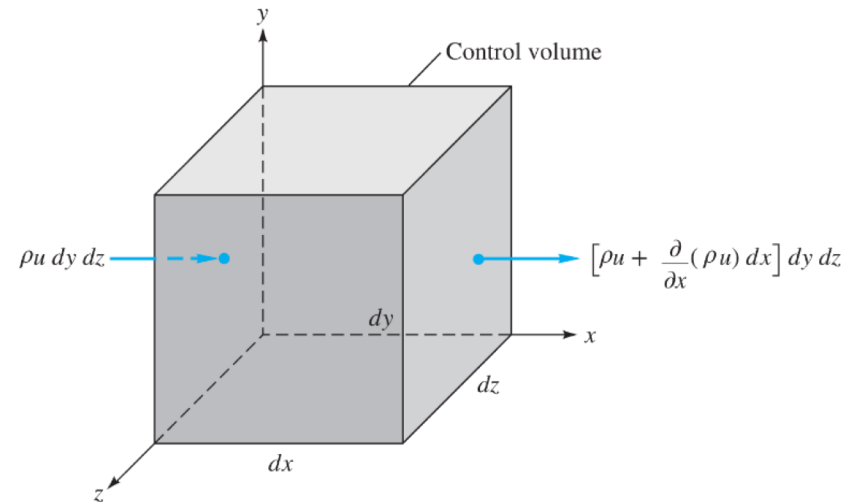
The Differential Equation of Mass Conservation

- The continuity equation in differential form (in Cartesian coordinates).

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z} = 0$$

- Cylindrical polar coordinates (r, θ, z)

$$\frac{\partial \rho}{\partial t} + \frac{1}{r} \frac{\partial(r \rho v_r)}{\partial r} + \frac{1}{r} \frac{\partial(\rho v_\theta)}{\partial \theta} + \frac{\partial(\rho v_z)}{\partial z} = 0$$



- Transformation of the coordinates, $r = \sqrt{x^2 + y^2}$ $\theta = \tan^{-1} \frac{y}{x}$ $z = z$
- For steady compressible flow,

$$\frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z} = 0$$

Incompressible flow

- For incompressible flow, the density change is negligible, $d\rho/dt = 0$ then regardless whether the flow is steady or unsteady.

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0 \quad \rightarrow \quad \nabla \cdot V = 0$$

- The continuity equation is always important and must always be satisfied for a rational analysis of a flow pattern.

Incompressible flow

Example:

An incompressible velocity field is given by

$$u = a(x^2 + y^2) \quad v \text{ unknown} \quad w = b$$

where a and b are constants.

What must the form of the velocity component v be?

Solution:

For incompressible flow, the continuity equation applies,

$$\frac{\partial [a(x^2 + y^2)]}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial b}{\partial z} = 0 \quad \text{or} \quad \frac{\partial v}{\partial y} = -2ax$$

Integrating partially with respect to y ,

$$v(x, y, z, t) = -2axy + f(x, z, t)$$

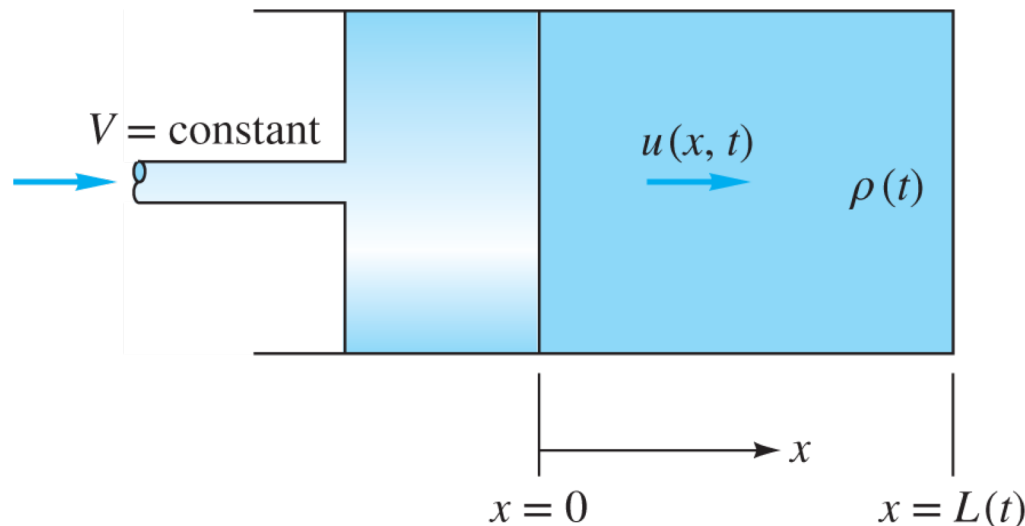
This is the only possible form for v that satisfies the continuity equation.

The Differential Equation of Mass Conservation

Example:

A piston compresses gas in a cylinder at constant speed $V = 12$ m/s. At one instant when the piston is at $L_0 = 0.15$ m away from the closed end of the cylinder, the gas density $\rho_0 = 18$ kg/m³. The gas velocity varies linearly from V at the piston to zero at the closed end. Find (1) an expression for the average gas density as a function of time, and (2) the rate of change of gas density at this instant.

(Assume the gas in the cylinder is unsteady, uniform, 1-D, and compressible)

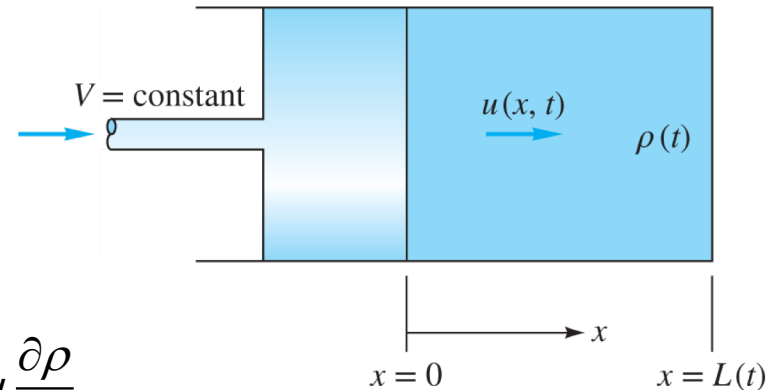


The Differential Equation of Mass Conservation

Solution:

Applying the continuity equation,

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u)}{\partial x} = 0 \quad \text{or} \quad \frac{\partial \rho}{\partial t} = -\frac{\partial(\rho u)}{\partial x} = -\rho \frac{\partial u}{\partial x} - u \frac{\partial \rho}{\partial x}$$



Note that $\frac{\partial \rho}{\partial x} = 0$ $u = V - \frac{V}{L}x$ $L = L_0 - Vt$ so $\frac{\partial \rho}{\partial t} = -\rho \frac{\partial u}{\partial x} = -\rho \frac{-V}{L_0 - Vt}$

Separate variables and integrate, $\int_{\rho_0}^{\rho} \frac{d\rho}{\rho} = \int_0^t \frac{V dt}{L_0 - Vt} \rightarrow \frac{\rho}{\rho_0} = \frac{L_0}{L_0 - Vt}$

At $t=0$, $\frac{\partial \rho}{\partial t} = \rho_0 \frac{V}{L_0 - Vt} = (18) \frac{12}{0.15} = 1440 \text{ kg/m}^3 \cdot \text{s}$

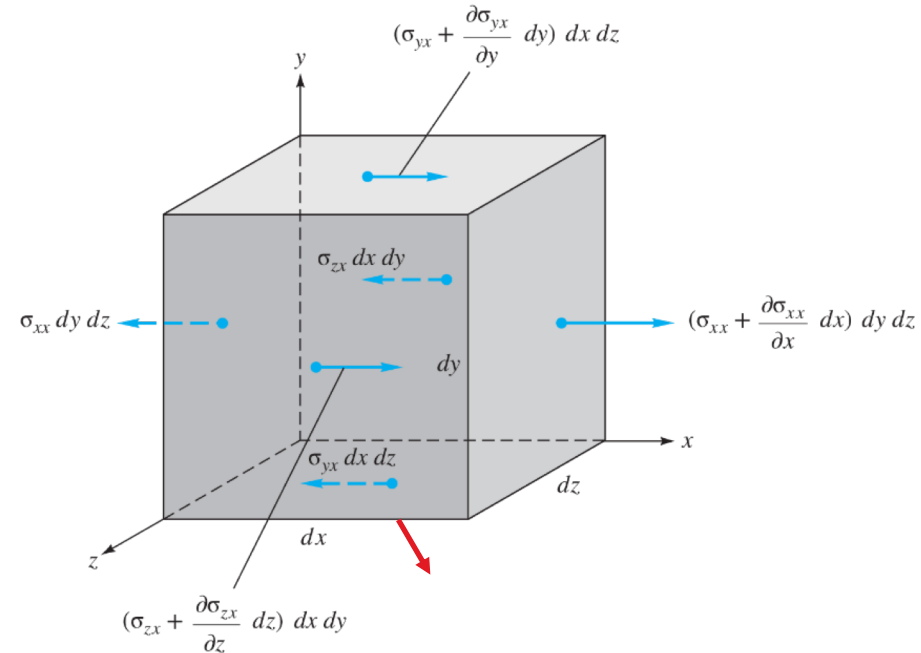
The Differential Equation of Linear Momentum

- The linear momentum for fixed CV and 1-D momentum flux,

$$\sum \mathbf{F} = \frac{d}{dt} \left(\int_{CV} \mathbf{v} \rho dV \right) + \sum_i (\dot{m}_i \mathbf{V}_i)_{out} - \sum_i (\dot{m}_i \mathbf{V}_i)_{in}$$

- Linear momentum changes in CV,

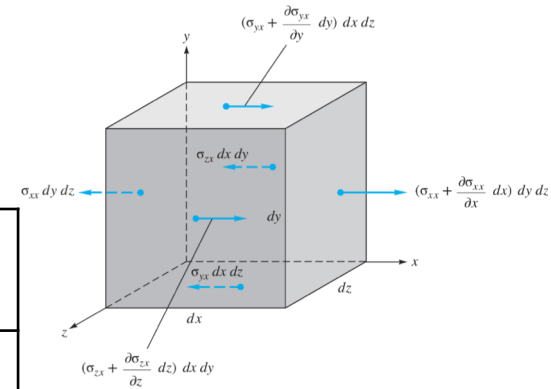
$$\frac{d}{dt} \left(\int_{CV} \mathbf{v} \rho dV \right) \approx \frac{\partial(\rho \mathbf{V})}{\partial t} dx dy dz$$



The Differential Equation of Linear Momentum

- Momentum flux terms in all six directions,

Face	Inlet momentum flow	Outlet momentum flow
x	$\rho u \mathbf{V} dy dz$	$\left[\rho u \mathbf{V} + \frac{\partial}{\partial x} (\rho u \mathbf{V}) dx \right] dy dz$
y	$\rho v \mathbf{V} dx dz$	$\left[\rho v \mathbf{V} + \frac{\partial}{\partial y} (\rho v \mathbf{V}) dy \right] dx dz$
z	$\rho w \mathbf{V} dx dy$	$\left[\rho w \mathbf{V} + \frac{\partial}{\partial z} (\rho w \mathbf{V}) dz \right] dx dy$



$$\rightarrow \sum \mathbf{F} = \left[\frac{\partial(\rho \mathbf{V})}{\partial t} + \frac{\partial(\rho u \mathbf{V})}{\partial x} + \frac{\partial(\rho v \mathbf{V})}{\partial y} + \frac{\partial(\rho w \mathbf{V})}{\partial z} \right] dx dy dz = \rho \frac{d\mathbf{V}}{dt} dx dy dz$$

Total Acceleration

Newtonian Viscous Flow: Navier-Stokes Equation

- The net force on the CV must be of differential size and proportional to the element volume. These forces include body forces and surface forces.

$$d\mathbf{F}_{\text{grav}} = \rho \mathbf{g} dx dy dz$$

Gravity force

$$\left(\frac{d\mathbf{F}}{dV} \right)_{\text{surf}} = -\nabla p + \left(\frac{d\mathbf{F}}{dV} \right)_{\text{vis}}$$

Normal and shear forces

where,

$$\left(\frac{d\mathbf{F}}{dV} \right)_{\text{vis}} = \mathbf{i} \left(\frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z} \right) + \mathbf{j} \left(\frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \tau_{yy}}{\partial y} + \frac{\partial \tau_{zy}}{\partial z} \right) + \mathbf{k} \left(\frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \tau_{zz}}{\partial z} \right)$$

More on Viscous Stresses

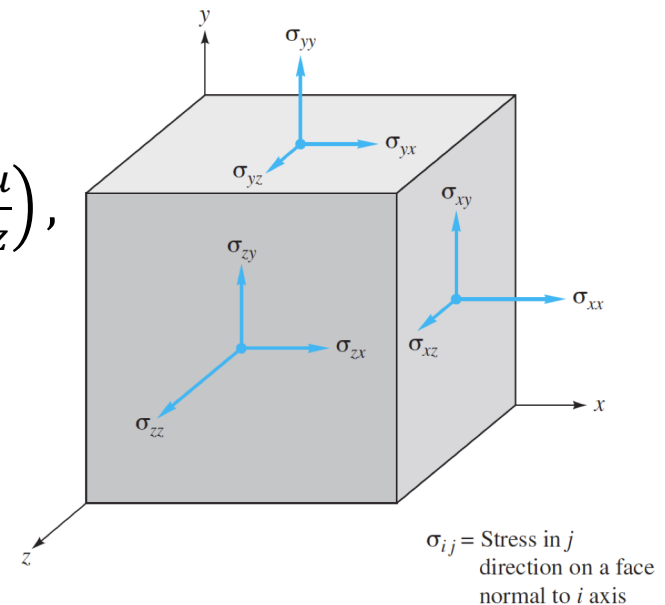
For a Newtonian fluid, the viscous stresses, τ_{ij} , are proportional to the element strain rates, and the coefficient of viscosity, μ .

For incompressible flow,

$$\begin{aligned}\tau_{xx} &= 2\mu \frac{\partial u}{\partial x}, \tau_{yy} = 2\mu \frac{\partial v}{\partial y}, \tau_{zz} = 2\mu \frac{\partial w}{\partial z} \\ \tau_{xy} &= \tau_{yx} = \mu \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right), \tau_{xz} = \tau_{zx} = \mu \left(\frac{\partial w}{\partial x} + \frac{\partial u}{\partial z} \right), \\ \tau_{yz} &= \tau_{zy} = \mu \left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} \right)\end{aligned}$$

where μ is the viscosity coefficient.

$$\sigma_{ij} = \begin{vmatrix} -p + \tau_{xx} & \tau_{yx} & \tau_{zx} \\ \tau_{xy} & -p + \tau_{yy} & \tau_{zy} \\ \tau_{xz} & \tau_{yz} & -p + \tau_{zz} \end{vmatrix}$$



σ_{ij} is stress, τ_{ij} viscous stress, and ϵ_{ij} is the strain rate $\left(\frac{1}{2} \left(\frac{u_i}{x_j} + \frac{u_j}{x_i} \right) \right)$.

They are nine-component tensor

Newtonian Viscous Flow: Navier-Stokes Equation

- **Incompressible Navier-Stokes Equations for a Newtonian fluid** with constant density and viscosity,

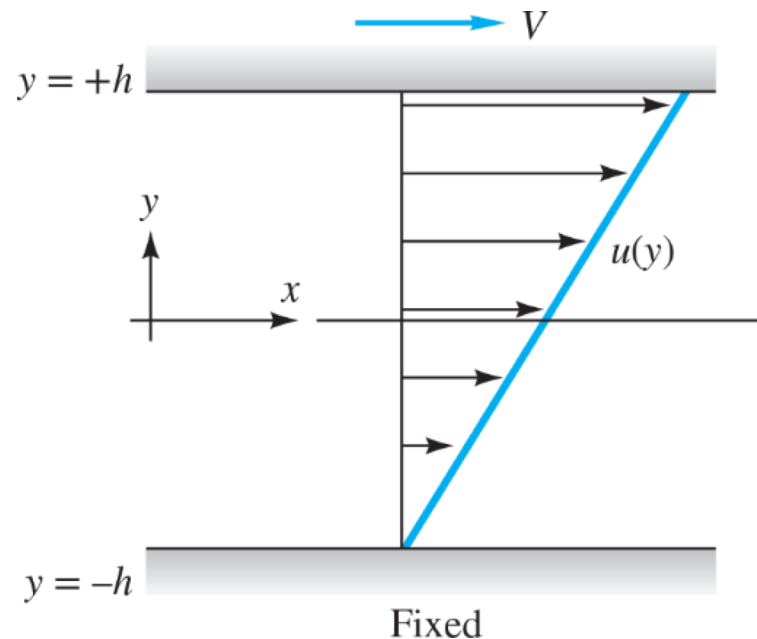
$$\begin{aligned} \rho \left(\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} \right) &= \rho g_x - \frac{\partial p}{\partial x} + \mu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) \\ \rho \left(\frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} \right) &= \rho g_y - \frac{\partial p}{\partial y} + \mu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} \right) \\ \rho \left(\frac{\partial w}{\partial t} + u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} \right) &= \rho g_z - \frac{\partial p}{\partial z} + \mu \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} \right) \end{aligned}$$

Mass × acceleration = gravity force + pressure force + viscous force

More generally , Navier-Stokes equations refer to mass, momentum, and energy conservation equations.

The Differential Equation of Linear Momentum

Example: (Couette Flow) Consider a 2-D incompressible viscous flow between parallel plates a distance $2h$ apart. We assume that the plates are very wide and long, so the flow is essentially axial, $u \neq 0$ but $v = w = 0$. The upper plate moves at velocity V . Neglect gravity effects, determine the velocity distribution between the plates.

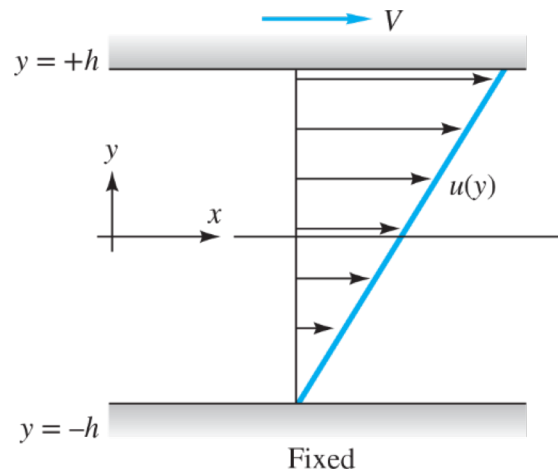


The Differential Equation of Linear Momentum

Solution:

This is a shear-driven flow. From the continuity eq.,

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0 \quad \rightarrow \quad \frac{\partial u}{\partial x} = 0 \quad \text{or} \quad u = u(y) \text{ only}$$



The Differential Equation of Linear Momentum

Solution (Con'd) :

There is a 'single' nonzero axial velocity component.

The flow is said to be steady and *fully developed*.

Applying the Navier-Stokes equations for 2-D flow, in the x direction,

$$\rho \left(u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} \right) = - \frac{\partial p}{\partial x} + \mu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right)$$

$$\Rightarrow \frac{\partial^2 u}{\partial y^2} = 0 \quad \text{or} \quad u = C_1 y + C_2$$

Boundary conditions at the upper and lower plates (no-slip condition) are:

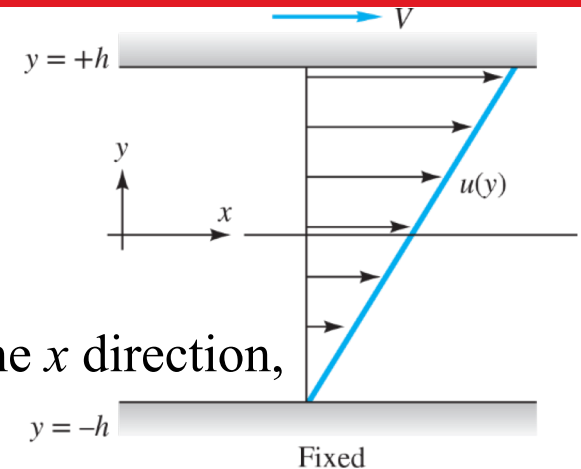
$$\text{At } y = h, \quad u = V = C_1 h + C_2$$

$$\text{At } y = -h, \quad u = 0 = C_1 (-h) + C_2$$

$$\Rightarrow C_1 = \frac{V}{2h} \quad \text{and} \quad C_2 = \frac{V}{2}$$

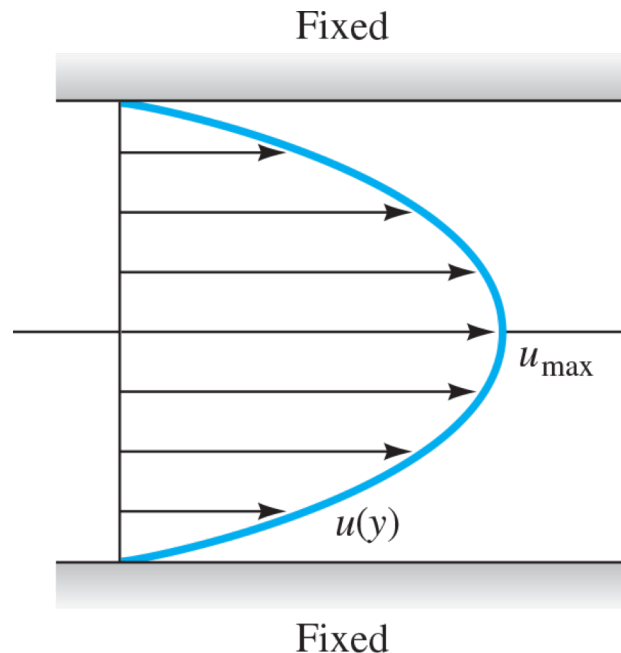
Velocity distribution:

$$u = \frac{V}{2h} y + \frac{V}{2}$$



The Differential Equation of Linear Momentum

Example: (Poiseuille Flow) Consider a 2-D incompressible viscous flow between parallel plates a distance $2h$ apart. Both plates are fixed, but pressure varies in x direction. If $u \neq 0$ but $v = w = 0$, neglect gravity effect, determine the velocity distribution between the plates.



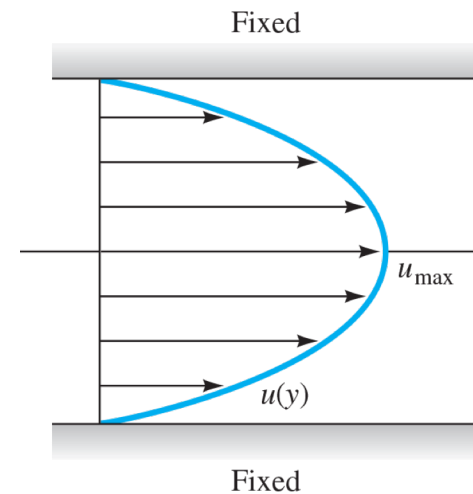
The Differential Equation of Linear Momentum

Solution:

This is a pressure-driven flow. From the continuity eq.,

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$$

$$\rightarrow \frac{\partial u}{\partial x} = 0 \quad \text{or} \quad u = u(y) \quad \text{only}$$



The Differential Equation of Linear Momentum

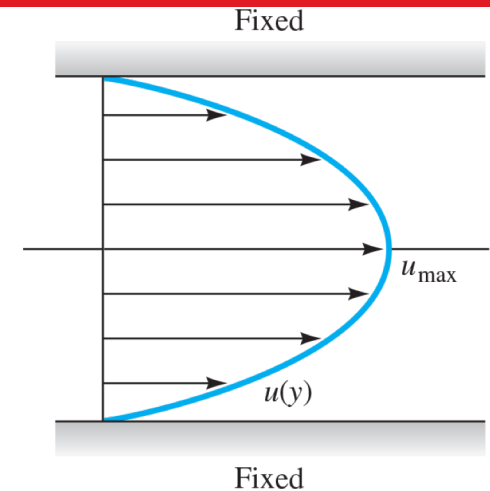
Solution (Cont'd):

Applying the N-S eqs. for 2-D flow, in the x direction,

$$\rho \left(u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} \right) = -\frac{\partial p}{\partial x} + \mu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right)$$

$\rightarrow \frac{dp}{dx} = \mu \frac{\partial^2 u}{\partial y^2}$ or $u = \frac{1}{\mu} \frac{dp}{dx} \frac{y^2}{2} + C_1 y + C_2$

Is dp/dx a constant?



Boundary conditions at the upper and lower plates (no-slip condition) are:

$$\text{At } y = h, \quad u = 0 = \frac{1}{\mu} \frac{dp}{dx} \frac{h^2}{2} + C_1 h + C_2$$

$$\text{At } y = -h, \quad u = 0 = \frac{1}{\mu} \frac{dp}{dx} \frac{h^2}{2} + C_1 (-h) + C_2$$

$\rightarrow C_1 = 0 \quad \text{and} \quad C_2 = -\frac{dp}{dx} \frac{h^2}{2\mu}$

Velocity distribution: $u = -\frac{dp}{dx} \frac{h^2}{2\mu} \left(1 - \frac{y^2}{h^2} \right)$ Maximum velocity: $u_{\max} = -\frac{dp}{dx} \frac{h^2}{2\mu}$

The Differential Equation of Energy

- The energy equation for fixed CV,

$$\dot{Q} - \dot{W}_s - \dot{W}_v = \frac{d}{dt} \left(\int_{CV} e \rho dV \right) + \int_{CS} \left(e + \frac{p}{\rho} \right) \rho (\mathbf{V} \cdot \mathbf{n}) dA$$

$\dot{W}_s = 0$ because no infinitesimal shaft work, but $\dot{W}_v \neq 0$ though small.

- By analogy with the linear momentum equation,

$$\dot{Q} - \dot{W}_v = \left[\frac{\partial(\rho e)}{\partial t} + \underbrace{\frac{\partial(\rho u \zeta)}{\partial x} + \frac{\partial(\rho v \zeta)}{\partial y} + \frac{\partial(\rho w \zeta)}{\partial z}}_{\text{Energy flux terms}} \right] dx dy dz$$

where $\zeta = e + p/\rho$

How can you get this?
Use energy flux terms on six subfaces.

Expand the last three terms by product rule and use the Continuity equation, (do the derivation at home)

$$\dot{Q} - \dot{W}_v = \left(\rho \frac{de}{dt} + V \cdot \nabla \rho + p \nabla \cdot V \right) dx dy dz \quad \dots \quad (\text{Eq.1})$$

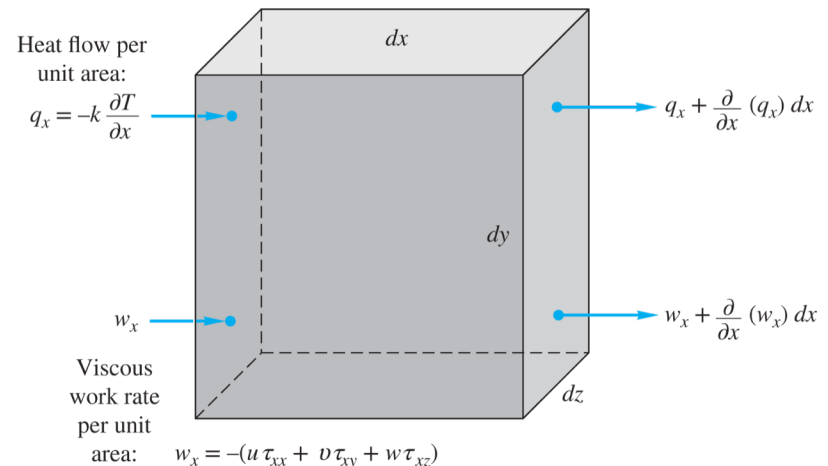
The Differential Equation of Energy

- **Fourier's law:** heat transfer per unit area is proportional to the gradient of temperature, where k is the **thermal conductivity**

$$q_x = -k \frac{\partial T}{\partial x} \quad q_y = -k \frac{\partial T}{\partial y} \quad q_z = -k \frac{\partial T}{\partial z}$$

$$\dot{Q} = - \left[\frac{\partial}{\partial x} (q_x) + \frac{\partial}{\partial y} (q_y) + \frac{\partial}{\partial z} (q_z) \right] dx dy dz \quad \dots\dots\dots (\text{Eq.2})$$

➔ $\dot{Q} = \nabla \cdot (k \nabla T) dx dy dz$
Total heat rate



The Differential Equation of Energy

- The net viscous work rate:

$$\dot{W}_v = - \left[\frac{\partial}{\partial x} (u\tau_{xx} + v\tau_{xy} + w\tau_{xz}) + \frac{\partial}{\partial y} (u\tau_{yx} + v\tau_{yy} + w\tau_{yz}) + \frac{\partial}{\partial z} (u\tau_{zx} + v\tau_{zy} + w\tau_{zz}) \right] dxdydz$$

$$\dot{W}_v = -\nabla \cdot (\mathbf{V} \cdot \boldsymbol{\tau}_{ij}) dxdydz \text{ where } \nabla \cdot (\mathbf{V} \cdot \boldsymbol{\tau}_{ij}) \equiv \mathbf{V} \cdot (\nabla \cdot \boldsymbol{\tau}_{ij}) + \Phi \dots\dots\dots (\text{Eq.3})$$

- Φ is the *viscous-dissipation function*. For a Newtonian incompressible viscous fluid, it is expressed as:

$$\Phi = \mu \left[2 \left(\frac{\partial u}{\partial x} \right)^2 + 2 \left(\frac{\partial v}{\partial y} \right)^2 + 2 \left(\frac{\partial w}{\partial z} \right)^2 + \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right)^2 + \left(\frac{\partial w}{\partial y} + \frac{\partial v}{\partial z} \right)^2 + \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} \right)^2 \right]$$

- Since all terms are quadratic, viscous dissipation is always positive, so a viscous flow always tends to lose its available energy due to dissipation, in accordance with the second law of thermodynamics.

The Differential Equation of Energy

- Plug Eq.2 and Eq.3 into Eq.1, then one form of differential energy equation:

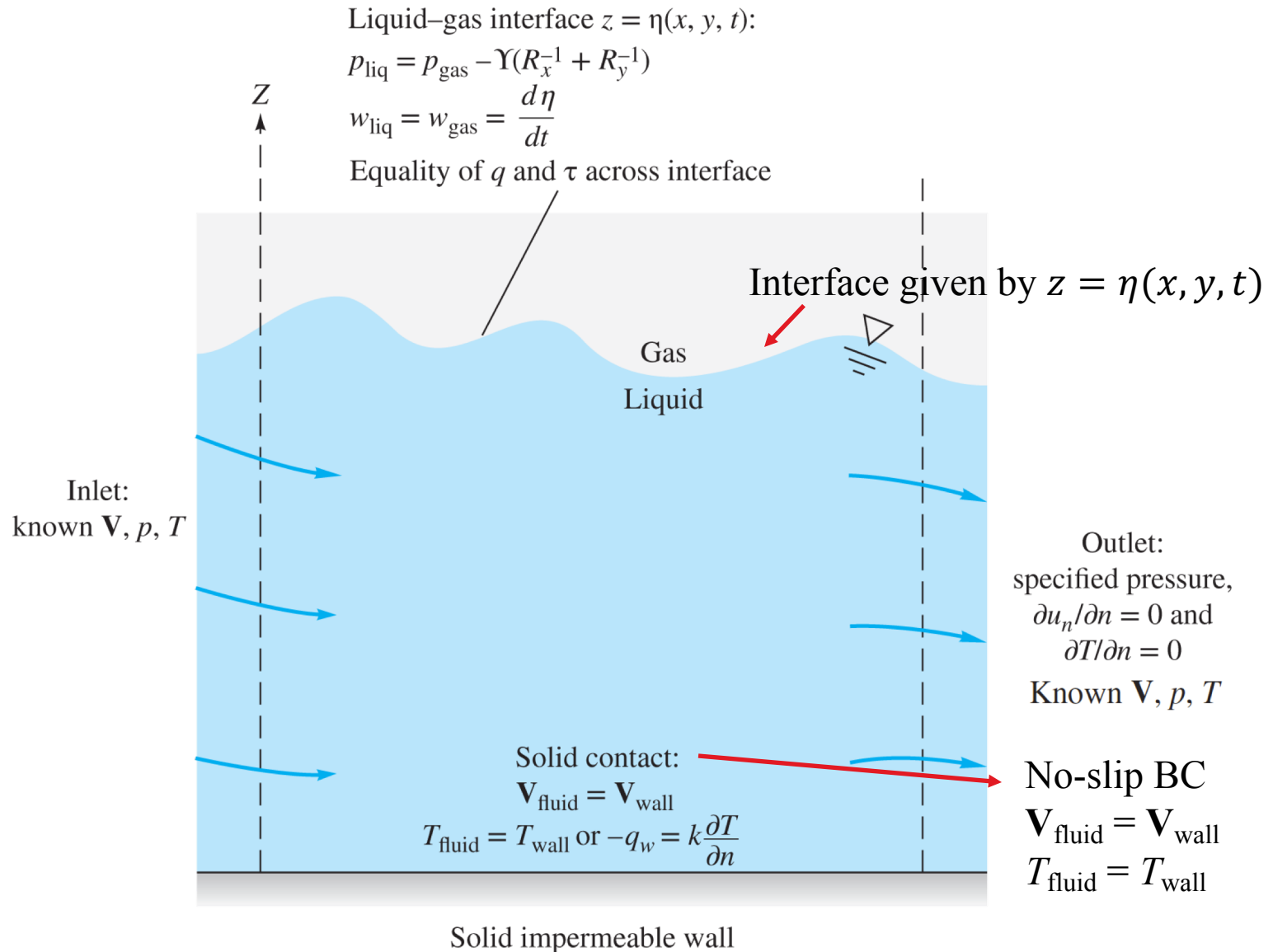
$$\rho \frac{de}{dt} + \mathbf{V} \cdot \nabla p + p \nabla \cdot \mathbf{V} = \nabla \cdot (k \nabla T) + \nabla \cdot (\mathbf{V} \cdot \boldsymbol{\tau}_{ij})$$

Boundary Conditions for the Basic Equations

- There are three most common types of boundaries encountered in fluid flow analysis: a solid wall, an inlet or outlet, and a liquid–gas interface.
- For incompressible flow with constant properties, continuity and momentum are independent of T .

At free surface, $p \approx p_a$ $w \approx \frac{\partial \eta}{\partial t}$

Boundary Conditions for the Basic Equations



-
- **Differential Form of Mass Conservation Equation**
 - **Differential Form of Momentum Conservation Equations**
 - **Differential Form of Energy Conservation Equation**
 - **Boundary Conditions**
 - **Stream Function**
 - **Vorticity, irrotationality**

The Stream Function

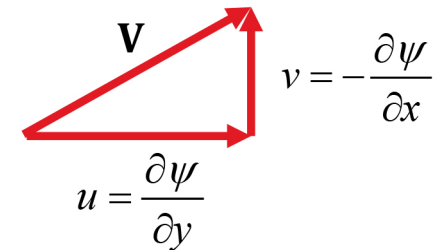
- Consider a steady, incompressible, plane (2-D) flow, the two velocity components u and v are related by the continuity equation as:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

ϕ : velocity potential
 ψ : stream function

- If we look for a function $\psi(x, y)$, called the stream function, such that it satisfies the continuity identically, they should be defined uniquely as:

$$u = \frac{\partial \psi}{\partial y} \quad v = -\frac{\partial \psi}{\partial x} \quad \text{or} \quad \mathbf{V} = \mathbf{i} \frac{\partial \psi}{\partial y} - \mathbf{j} \frac{\partial \psi}{\partial x}$$



- Stream function is different from velocity potential function of ϕ , where

$$u = \frac{\partial \phi}{\partial x} \quad v = \frac{\partial \phi}{\partial y} \quad \text{or} \quad \mathbf{V} = \mathbf{i} \frac{\partial \phi}{\partial x} + \mathbf{j} \frac{\partial \phi}{\partial y}$$

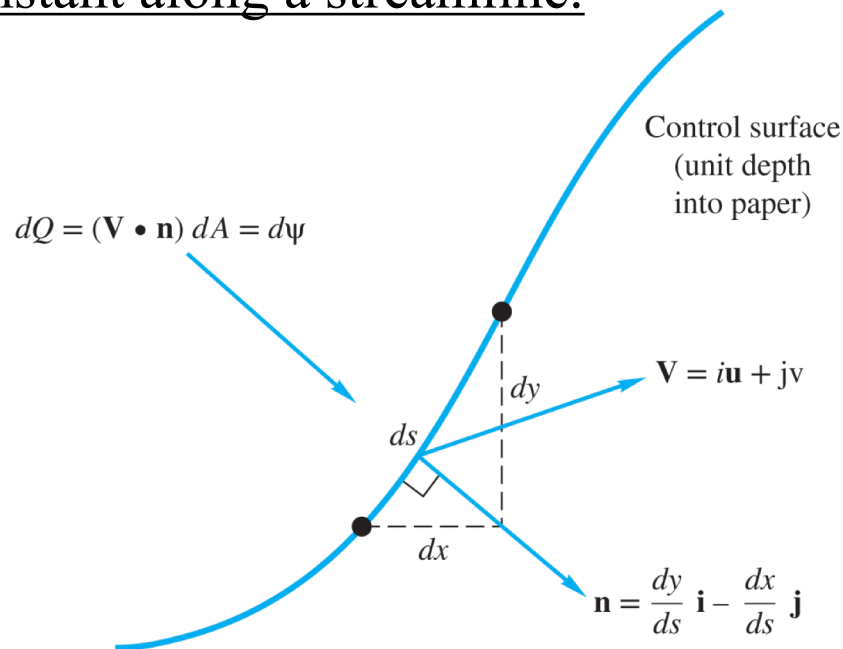
Geometric Interpretation of the Stream Function

- **Lines of constant ψ are streamlines of the flow.**

For a streamline,

$$\frac{dx}{u} = \frac{dy}{v} \quad \text{or} \quad udy - vdx = 0 \quad \rightarrow \quad \frac{\partial \psi}{\partial x} dx + \frac{\partial \psi}{\partial y} dy = 0 = d\psi$$

ψ is constant along a streamline.



Geometric Interpretation of the Stream Function

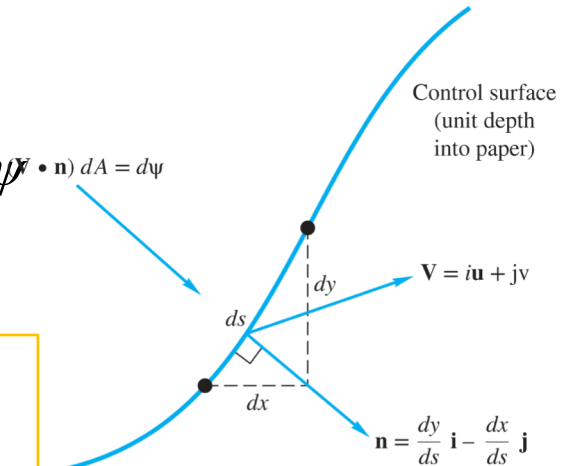
- The change in the value of ψ is related to the volume flow rate of flow.

Consider control surface of $ds \times 1$

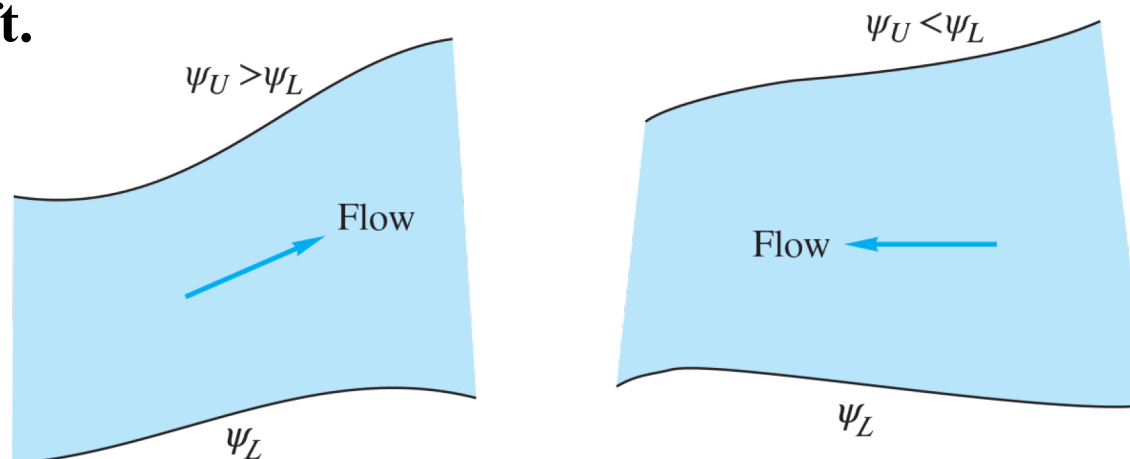
$$dQ = (\mathbf{V} \cdot \mathbf{n}) dA = \left(\mathbf{i} \frac{\partial \psi}{\partial y} - \mathbf{j} \frac{\partial \psi}{\partial x} \right) \cdot \left(\mathbf{i} \frac{dy}{ds} - \mathbf{j} \frac{dx}{ds} \right) ds (1) = d\psi$$

$$\rightarrow Q_{1-2} = \psi_2 - \psi_1$$

The volume flow between any two streamlines in the flow field is equal to the change in stream function between those streamlines. What about the direction?



- The flow is to the right if ψ_U is greater than ψ_L , otherwise the flow is to the left.



The Stream Function

Example: If a stream function exists for a velocity field,

$$u = a(x^2 - y^2) \quad v = -2axy \quad w = 0$$

find, plot, and interpret the velocity field.

The Stream Function

Solution: Assume incompressible and 2-D flow.

Step 1: To confirm the velocity field satisfies continuity,

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = \frac{\partial}{\partial x} [a(x^2 - y^2)] + \frac{\partial}{\partial y} (-2axy) = 2ax - 2ax = 0$$

Step 2: To find ψ

$$u = \frac{\partial \psi}{\partial y} = a(x^2 - y^2) \quad (1)$$

$$v = -2axy \quad (2)$$

Integrate (1) with respect to y , $\psi = ax^2 y - \frac{ay^3}{3} + f(x) \quad (3)$

Differentiate Eq. (3) with respect to x , $\frac{\partial \psi}{\partial x} = 2axy + f'(x) = 2axy$

Therefore, $f'(x) = 0$, or $f = \text{const.}$, the stream function is $\psi = a \left(x^2 y - \frac{y^3}{3} \right) + C$

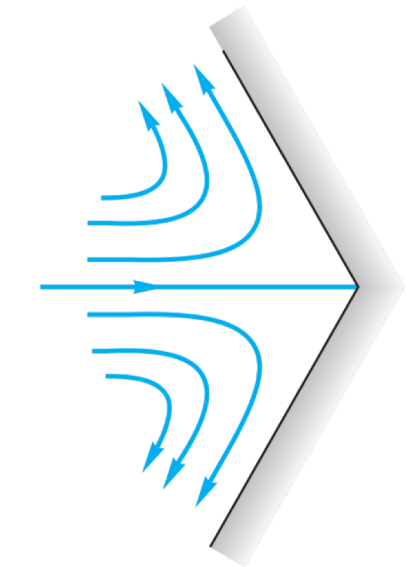
The Stream Function

Solution (cont'd): To plot this, set $C = 0$ for convenience

$$3x^2y - y^3 = \frac{3\psi}{a}$$

For constant value of ψ , the results are shown.

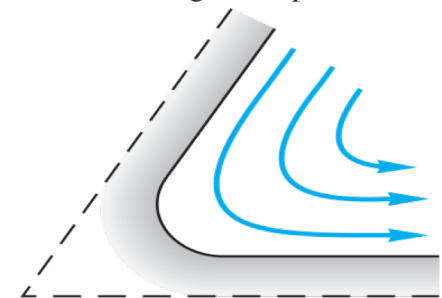
To interpret,



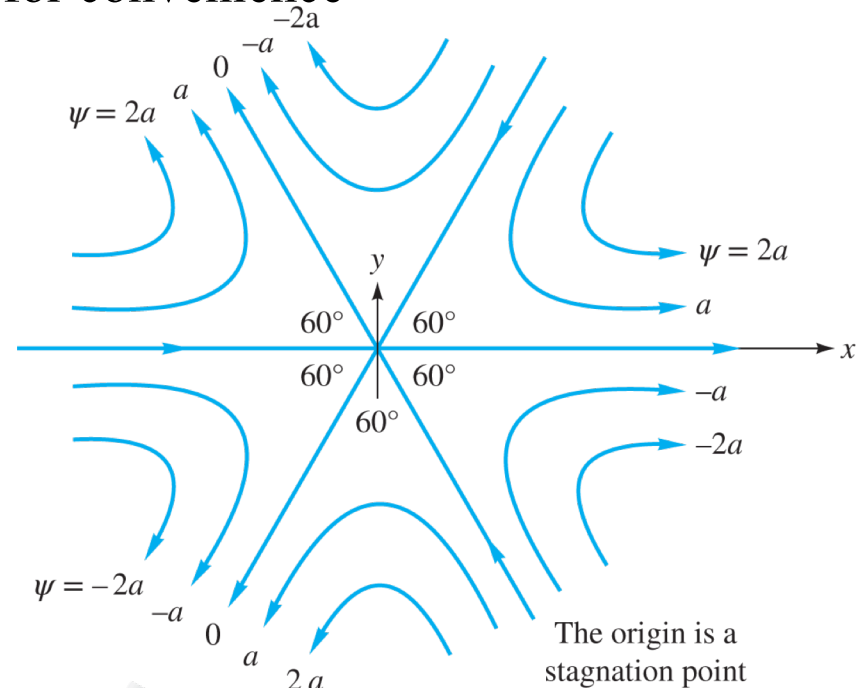
Incoming stream impinging against a 120° corner



Flow around a 60° corner



Flow around a rounded 60° corner



Vorticity

- The vorticity or *curl* $\mathbf{V}$ ($w = 0$) is:

$$\zeta = \nabla \times \mathbf{V} = \text{curl } \mathbf{V} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ u & v & w \end{vmatrix} = \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \mathbf{k} = -\mathbf{k} \left(\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} \right) = -\mathbf{k} \nabla^2 \psi$$

- Physical meaning of vorticity: the average rate of counter-clockwise turning of a fluid element (multiplied by 2)

- If the flow is **irrotational**, $\text{curl } \mathbf{V} = 0 \rightarrow \frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} = 0$
Laplace's equation

Vorticity and Irrotationality

- The angular velocity ω_z is defined as the average rate of the counter-clockwise turning of the two lines:

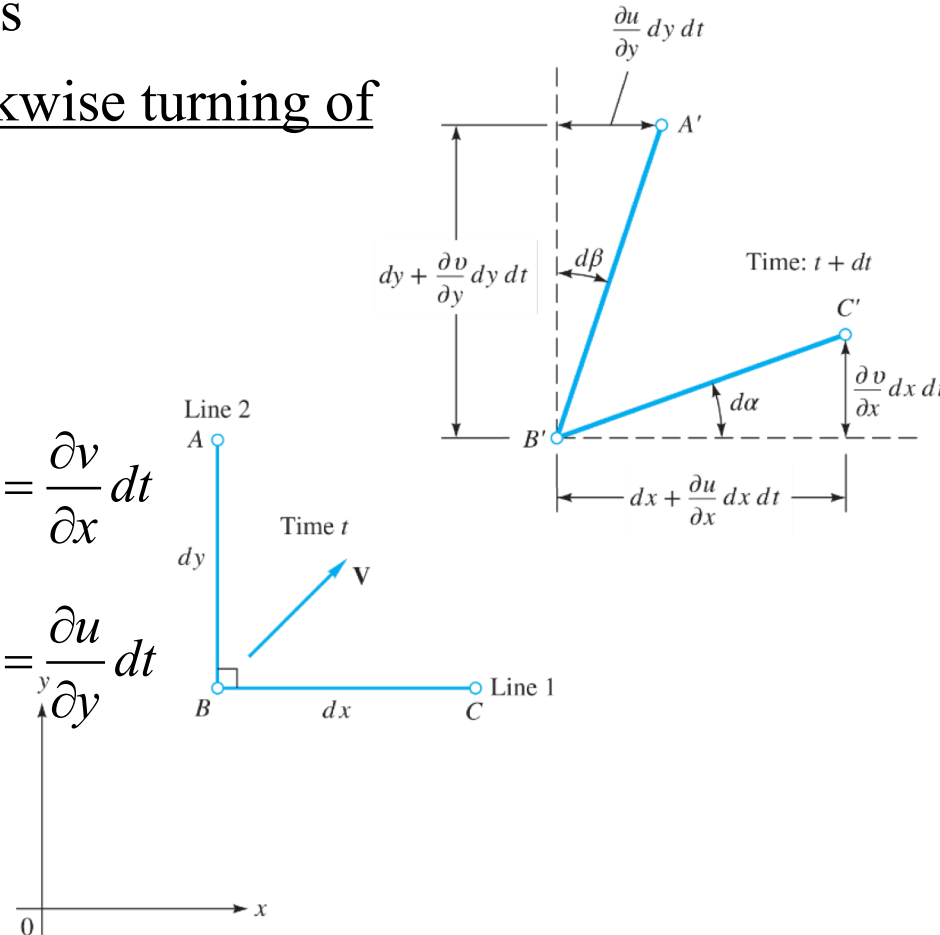
$$\omega_z = \frac{1}{2} \left(\frac{\partial \alpha}{\partial t} - \frac{\partial \beta}{\partial t} \right)$$

where $d\alpha = \lim_{dt \rightarrow 0} \left[\tan^{-1} \frac{(\partial v / \partial x) dx dt}{dx + (\partial u / \partial x) dx dt} \right] = \frac{\partial v}{\partial x} dt$

$$d\beta = \lim_{dt \rightarrow 0} \left[\tan^{-1} \frac{(\partial u / \partial y) dy dt}{dy + (\partial v / \partial y) dy dt} \right] = \frac{\partial u}{\partial y} dt$$

The angular velocity is,

$$\omega_z = \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right)$$



Vorticity and Irrotationality

In 3-D flow,

$$\omega_x = \frac{1}{2} \left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \right) \quad \omega_y = \frac{1}{2} \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \right) \quad \omega_z = \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right)$$

The vector $\boldsymbol{\omega} = \mathbf{i}\omega_x + \mathbf{j}\omega_y + \mathbf{k}\omega_z$

$$\rightarrow \boldsymbol{\omega} = \frac{1}{2} \text{curl } \mathbf{V} = \frac{1}{2} \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ u & v & w \end{vmatrix}$$

Vorticity and Irrotationality

- The *vorticity* is defined as a vector twice as large.

$$\boldsymbol{\zeta} = 2\boldsymbol{\omega} = \text{curl}\mathbf{V}$$

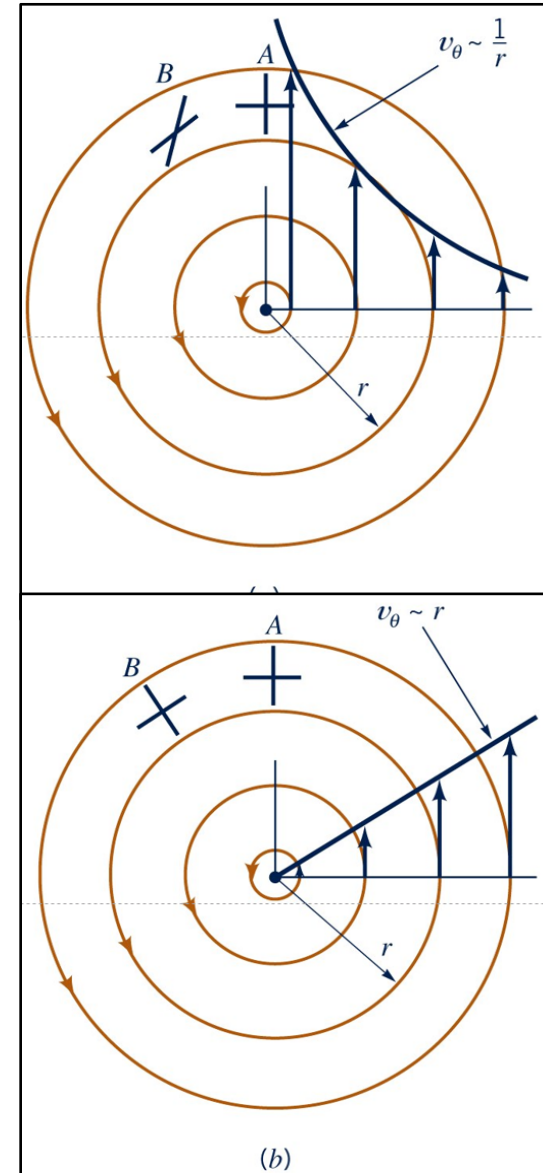
- Flows have negligible or zero vorticity are called **irrotational**:

$$\text{curl}\mathbf{V} \equiv 0$$

- Rotating flow is NOT equivalent to rotational flow.

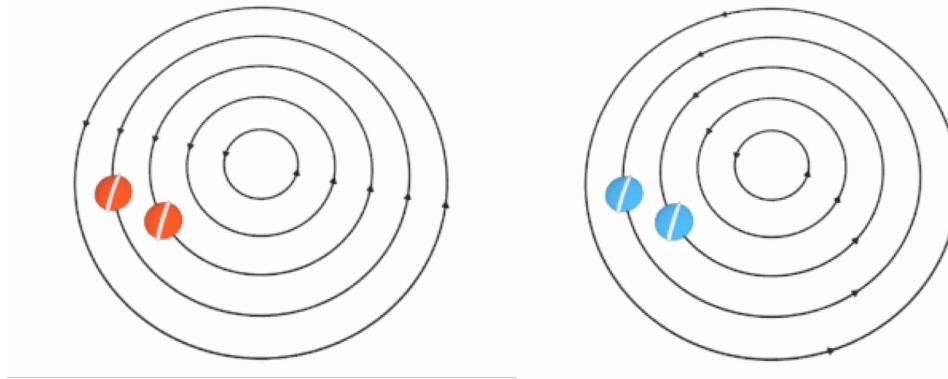
Free Vortex vs. Forced Vortex

- Free vortex (irrotational vortex)
 - E.g., $\bar{V} = (v_r = 0, v_\theta = \frac{c}{r})$, $(\nabla \times \bar{V})_z = \frac{1}{r} \left(\frac{\partial(rv_\theta)}{\partial r} - \frac{\partial v_r}{\partial \theta} \right) = 0$
 - Average rate of counter-clockwise turning is zero
 - Orientation of fluid element remains the same
- Forced vortex (rotational vortex, rigid-body vortex)
 - E.g., $\bar{V} = (v_r = 0, v_\theta = cr)$, $(\nabla \times \bar{V})_z = \frac{1}{r} \left(\frac{\partial(rv_\theta)}{\partial r} - \frac{\partial v_r}{\partial \theta} \right) = 2c$
 - Average rate of counter-clockwise turning is NOT zero
 - Orientation of fluid element continuously rotates



Free Vortex vs. Forced Vortex

- Identify which one is free vortex and forced vortex.



- Concepts of free vortex and forced vortex are used for analysis of wing-tip vortex, hurricane, plughole vortex, etc.



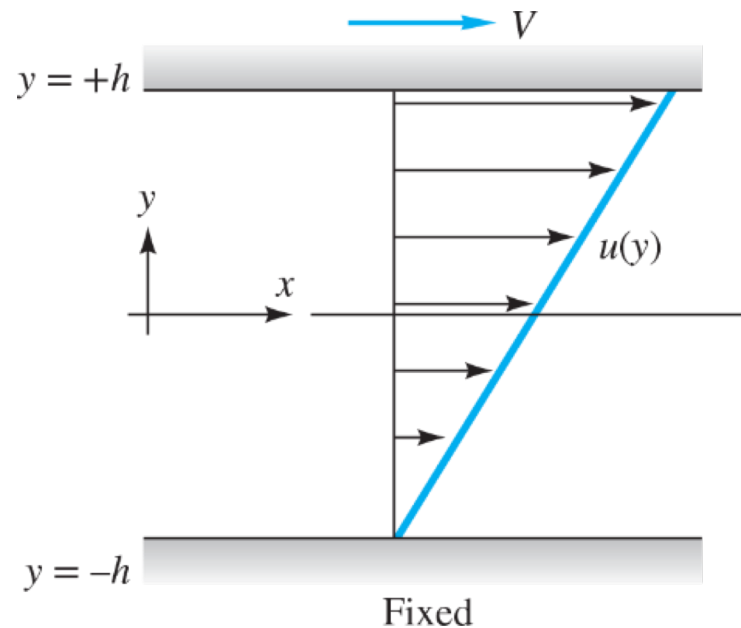
https://en.wikipedia.org/wiki/Vortex#/media/File:Airplane_vortex_edit.jpg

<https://earthobservatory.nasa.gov/images/147166/typhoon-bavi-approaches-korean-peninsula>

<https://www.gettyimages.fi/detail/photo/vortex-of-water-going-down-plughole-underwater-view-royalty-free-image/200286107-001>

Vorticity and Irrotationality

Example: Consider two plane flows: (1) a Couette flow, $u = ay$, $v = 0$, where a is a constant. (2) a “potential vortex” flow, which is a useful model for phenomena such as tornadoes and whirlpools, $v_r = 0$, $v_\theta = K/r$, where K is a constant. Check if above flows are rotational.



Vorticity and Irrotationality

Solution:

(1) For the Couette flow, $\omega_z = \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) = -\frac{1}{2} a$

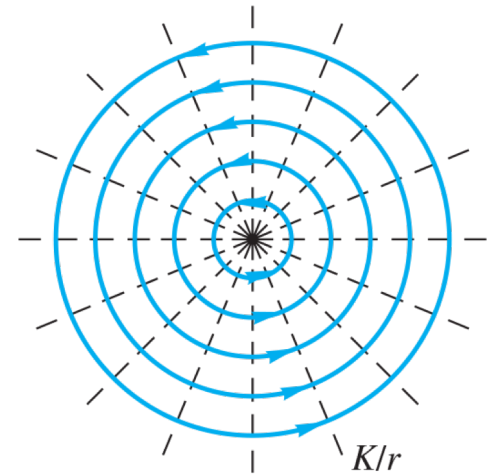
The angular velocity is constant; therefore it is rotational everywhere in the flow.

(2) The “potential vortex” flow is given in polar coordinates, find ω_z from Appendix D,

$$\omega_z = \frac{1}{2} \left(\frac{1}{r} \frac{\partial (rv_\theta)}{\partial r} - \frac{1}{r} \frac{\partial v_r}{\partial \theta} \right) = 0$$

Surprising results?

Couette flow



Potential vortex

Frictionless Irrotational Flows

- From vector analysis, a vector with zero curl must be the gradient of a scalar function,

$$\text{If } \text{curl } \mathbf{V} = \nabla \times \mathbf{V} \equiv 0 \quad \text{then } \mathbf{V} = \nabla \phi$$

where $\phi = \phi(x, y, z, t)$ is called the velocity potential function. So,

$$u = \frac{\partial \phi}{\partial x} \quad v = \frac{\partial \phi}{\partial y} \quad w = \frac{\partial \phi}{\partial z}$$

Lines of constant ϕ are called the potential lines of the flow.

Frictionless Irrotational Flows

- If a plan flow is irrotational, and ψ and ϕ both exist, the streamlines and potential lines are everywhere mutually perpendicular except at a stagnation point. For incompressible flow in the xy plane,

$$u = \frac{\partial \psi}{\partial y} = \frac{\partial \phi}{\partial x} \quad v = -\frac{\partial \psi}{\partial x} = \frac{\partial \phi}{\partial y}$$

- The above equations are also called *the Cauchy--Riemann equations*. So ψ and ϕ satisfy Laplace's equation.

Inviscid, Irrotational Flow and Velocity Potential

- Consider the momentum eq. of inviscid (frictionless) flow,

$$\rho \frac{D\bar{V}}{Dt} = -\nabla p + \rho \bar{g}, \text{ LHS becomes}$$

$$\frac{D\bar{V}}{Dt} = \frac{\partial \bar{V}}{\partial t} + \bar{V} \cdot \nabla \bar{V} = \frac{\partial \bar{V}}{\partial t} + \nabla \left(\frac{1}{2} V^2 \right) + \nabla \times \bar{V} \times \bar{V} = \frac{\partial \bar{V}}{\partial t} + \nabla \left(\frac{1}{2} V^2 \right) + \bar{\zeta} \times \bar{V}$$

$$\frac{\partial \bar{V}}{\partial t} + \nabla \left(\frac{1}{2} V^2 \right) + \bar{\zeta} \times \bar{V} + \frac{1}{\rho} \nabla p - \bar{g} = 0$$

- For an arbitrary displacement $d\bar{r}$,

$$\left[\frac{\partial \bar{V}}{\partial t} + \nabla \left(\frac{1}{2} V^2 \right) + \bar{\zeta} \times \bar{V} + \frac{1}{\rho} \nabla p - \bar{g} \right] \cdot d\bar{r} = 0$$

Inviscid, Irrotational Flow and Velocity Potential

$$\left[\frac{\partial \bar{V}}{\partial t} + \nabla \left(\frac{1}{2} V^2 \right) + \bar{\zeta} \times \bar{V} + \frac{1}{\rho} \nabla p - \bar{g} \right] \cdot d\bar{r} = 0$$

- If $(\bar{\zeta} \times \bar{V}) \cdot d\bar{r} = 0$, then we can obtain the Bernoulli equation.

(1) $d\bar{r} \parallel \bar{V}$: $d\bar{r}$ is placed on a streamline

For inviscid unsteady flow, the Bernoulli equation between any two points 1 and 2 along the streamline is written as

$$\int_1^2 \frac{\partial V}{\partial t} ds + \int_1^2 \frac{dp}{\rho} + \frac{1}{2} (V_2^2 - V_1^2) + g(z_2 - z_1) = 0$$

(2) $\bar{\zeta} = 0$: irrotational flow; the direction of $d\bar{r}$ does not matter

For incompressible, steady, inviscid and irrotational flow,

$$p + \frac{1}{2} \rho V^2 + \rho g z = \text{const} \text{ is valid } \underline{\text{anywhere}}$$

Inviscid, Irrotational Flow and Velocity Potential

- For irrotational flow, we can define a function ϕ such that $\bar{V} = \nabla\phi$

$$\nabla \times \nabla\phi = 0 \text{ for any scalar } \phi, \text{ so } \nabla \times \nabla\phi = \nabla \times \bar{V} = 0$$

- Velocity potential function

Irrotational flow is equivalent to potential flow

Continuity equation becomes Laplace equation

$$\nabla \cdot \bar{V} = \nabla \cdot (\nabla\phi) = \nabla^2\phi = 0$$

- Potential function ϕ is orthogonal to stream function ψ

Consider 2D flow

$$u = \frac{\partial\psi}{\partial y} = \frac{\partial\phi}{\partial x}, \quad v = -\frac{\partial\psi}{\partial x} = \frac{\partial\phi}{\partial y}$$

$$\text{For a constant } \phi \text{ line, } d\phi = 0 = \frac{\partial\phi}{\partial x}dx + \frac{\partial\phi}{\partial y}dy = udx + vdy$$

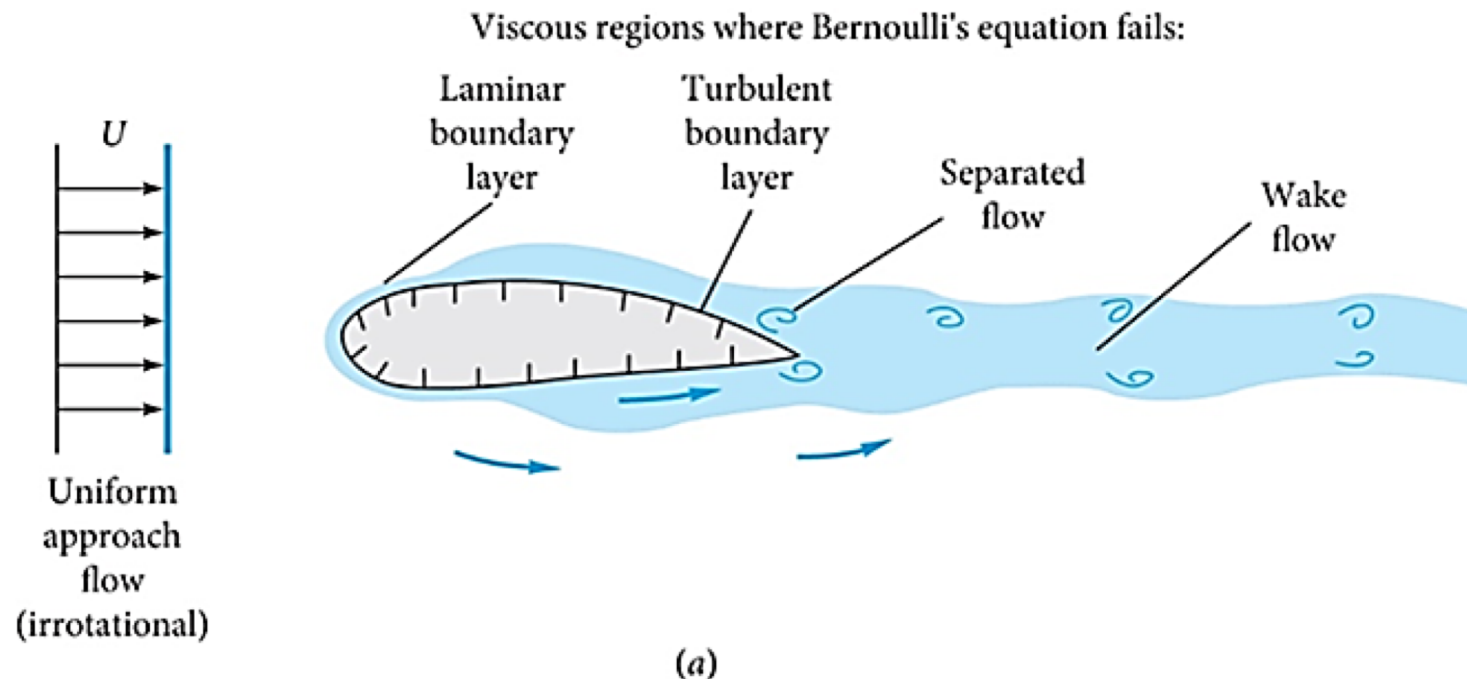
$$\text{For a constant } \psi \text{ line, } d\psi = 0 = \frac{\partial\psi}{\partial x}dx + \frac{\partial\psi}{\partial y}dy = -vdx + udy$$

$$\left. \frac{dy}{dx} \right|_{\phi} = -\frac{u}{v} = -\frac{1}{\left. \frac{dy}{dx} \right|_{\psi}}$$

Inviscid, Irrotational Flow and Velocity Potential

- In which region the assumption of potential flow is valid?

Incompressible flow outside the boundary layer



Frictionless Irrotational Flows

Example:

If a velocity potential exists for a velocity field

$$u = a(x^2 - y^2) \quad v = -2axy \quad w = 0$$

find it, plot it, and compare it with the previous example.

Frictionless Irrotational Flows

Solution: Assume incompressible and 2-D flow.

Step 1: To confirm the flow is irrotational,

$$\text{curl } \mathbf{V} = 2\omega_z = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = \frac{\partial}{\partial x}(-2axy) - \frac{\partial}{\partial y}(ax^2 - ay^2) = 2ax - 2ax = 0$$

Step 2: To find ϕ

$$u = \frac{\partial \phi}{\partial x} = a(x^2 - y^2) \quad (1)$$

$$v = -2axy \quad (2)$$

Integrate (1) with respect to x , $\phi = \frac{ax^3}{3} - axy^2 + f(y) \quad (3)$

Differentiate Eq. (3) with respect to y , $\frac{\partial \phi}{\partial y} = -2axy + f'(y) = -2axy$

Therefore, $f'(y) = 0$, or $f = \text{constant}$, the velocity potential is $\phi = a\left(\frac{x^3}{3} - xy^2\right) + C$

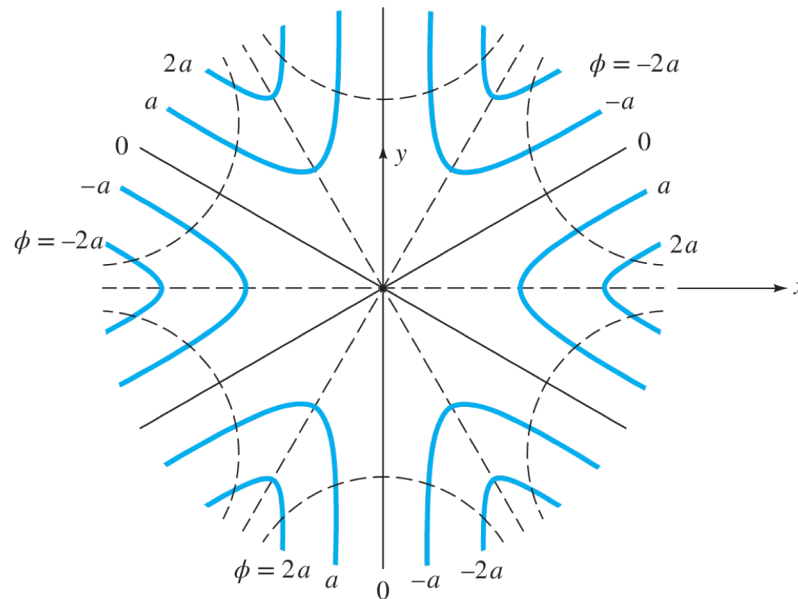
Frictionless Irrotational Flows

Example (cont.): To plot this, set $C = 0$ for convenience

$$3xy^2 - x^3 = \frac{3\phi}{a}$$

For constant value of ϕ , the results are shown.

The ϕ and ψ lines are everywhere perpendicular except at the origin, a stagnation point, where they are 30° apart.



- The differential relation for the conservation of mass is often called the continuity equation. For incompressible flow,

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$$

- The differential equations for linear momentum in Newtonian and viscous flow is known as the Navier-Stokes equations.

$$\rho \mathbf{g} - \nabla p + \nabla \cdot \boldsymbol{\tau}_{ij} = \rho \frac{d\mathbf{V}}{dt}$$

- The differential equation for energy for incompressible flow,

$$\rho c_v \left(\frac{\partial T}{\partial t} + u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} + w \frac{\partial T}{\partial z} \right) = k \nabla^2 T + \Phi$$

- There are three most common types of boundaries encountered in fluid flow analysis: a solid wall, an inlet or outlet, and a liquid--gas interface.
- The stream function is defined as $u = \frac{\partial \psi}{\partial y}$ $v = -\frac{\partial \psi}{\partial x}$
- The vorticity or curl $\mathbf{V}$ is $\text{curl } \mathbf{V} = -\mathbf{k} \nabla^2 \psi$
- For irrotational flow, $\text{curl } \mathbf{V} = 0$